

MPGI SPORTS

APEX 2026 Inter-College Championship

Resend Email System

Complete Technical Documentation

SENDER	MPGI Sports <noreply@mpgisports.in>
DOMAIN	mpgisports.in (Resend Verified '')
API	Resend Transactional Email API v6.22.1
BRANCH	ankithome (GitHub — production ready)
STATUS	LIVE & PRODUCTION READY '

01 Overview — System Kya Karta Hai?

Jab bhi koi student mpgisports.in par apne game ke liye registration karta hai, system AUTOMATICALLY us student aur uski puri team ko ek official confirmation email bhejta hai.

- Email sirf ek click me bhejta hai — registration ke time koi delay nahi
- Email kabhi registration fail nahi karega — agar email na jaye, registration phir bhi successful rahega
- Email me directly PDF Pass download hota hai — koi website par login karne ki zarurat nahi

02 Email Kahan Se Aata Hai?

From (Sender)	MPGI Sports <noreply@mpgisports.in>
Domain	mpgisports.in (Resend par verified)
API Provider	Resend Transactional Email SDK v6.22.1
RESEND_API_KEY	Environment Variable (never hardcoded)
RESEND_FROM_EMAIL	Environment Variable (never hardcoded)
APP_URL	https://mpgisports.in

03 Email Kab Bheja Jata Hai — Exact Flow

```
$` Student Form Submit karta hai
```

```
!`
```

```
$a Backend DB me Registration Save hoti hai
```

```
!`
```

```
$b Student ko Screen par Receipt IMMEDIATELY dikhayi deti hai (< 0.1 sec)
```

```
!` (Iske baad background me — user ko wait nahi karna)
```

```
$c triggerAsyncRegistrationEmail() function call hota hai (setImmediate)
```

```
!`
```

```
$d Resend API! Email dispatch hota hai
```

```
!`
```

```
$e Captain + Team Members ke inbox me email aata hai '
```

04 Kiske Paas Email Jaata Hai — Dynamic Routing

A. Individual / Solo Events (Chess, Badminton Singles, TT Singles, 100m, etc.)

- Sirf 1 email jaata hai — registered player ke email par
- Email me: Player naam, sport, college, receipt ID, PDF pass button

B. Doubles / Duo Events (Badminton Doubles, Table Tennis Doubles)

- 2 emails jaate hain — Player 1 aur Player 2 dono ke email par

- Deduplication: Agar dono ne same email likha ho, sirf 1 email jaata hai
- Email me: Dono players naam, Captain badge, receipt ID, PDF pass button

C. Team Events (Cricket, Football, Basketball, Volleyball, Kabaddi, Kho-Kho, etc.)

- Captain + sare team members ke email par email jaata hai
- Empty ya invalid emails automatically filter ho jaate hain
- Duplicate emails automatically remove ho jaate hain (deduplication)
- Email me: Captain badge, puri roster table, har player naam + roll no, PDF button

05 Email Ke Andar Kya-Kya Hota Hai?

Section	Content
Header	MPGI SPORTS + APEX 2026 Inter-College Championship branding
Success Alert	Registration Successful badge with CONFIRMED status (green)
Receipt Box	Unique Receipt ID (REC-APEX-XXXX) in gold highlighted box
Details Grid	Participant, Sport, Category, Captain, College, Team, Fee, Payment Ref
Roster Table	Complete squad with Sr.No, Name, Captain badge, Enrollment No. (Team/Doubles)
Download CTA	Download Registration Pass (PDF) button — direct PDF download
Instructions	College ID card, 45 min early reporting, all players for kit verification
Footer	MPGI Sports Committee mpgisports.in support@mpgisports.in

06 PDF Pass Download — Kaise Kaam Karta Hai?

\$` Student email me "Download Registration Pass (PDF)" button dabata hai !"

\$a Button!' Secure backend URL par redirect:

<https://mpgisports.in/api/public/registration-pass/REC-APEX-XXXX>

!"

\$b Backend database se REAL-TIME data uthata hai

!"

\$c Server-side jsPDF se A4 PDF pass on-the-fly generate hota hai

!"

\$d Browser me bina kisi page ke — PDF file DIRECT DOWNLOAD ho jaati hai

- Koi login required nahi — koi bhi team member apna pass download kar sakta hai
- Content-Disposition: attachment header se direct download start hoti hai

07 Duplicate Email Prevention — Idempotency

Level 1 — In-Flight Lock (Memory):

- Ek Set<receiptId> memory me active jobs track karta hai
- Agar same receipt ke liye 2 triggers aayein, doosra turant ignore ho jaata hai

Level 2 — Database Status Check:

- email_status = "sent" already ho to dobara email nahi jaata
- Razorpay webhook retry ya page reload se kabhi duplicate email nahi jaayega

08 Email Failure Isolation — Registration Safe Rahegi

Component	Status if Email Fails
Registration	' Successful rahegi — koi rollback nahi
Payment	' Successful rahegi — koi refund nahi
Receipt Screen	' User ko dikhayi degi
PDF Pass	' Download hoga via backend route
Email	'L Fail hogi — DB me "failed" status + error log

Database me email_status = "failed" aur email_error column me error message save ho jaata hai — sirf log ke liye.

09 Database Columns — Auto-Added on Server Start

Column	Type	Purpose
email_status	TEXT	pending / sent / failed / skipped
email_error	TEXT	Error message (if failed)
email_sent_at	TIMESTAMP	Successful delivery timestamp

In columns ke liye ALTER TABLE college_registrations ADD COLUMN IF NOT EXISTS ... run hota hai — koi existing data delete nahi hota.

10 Files Jo Implement Kiye Gaye

File	Kaam
server/services/emailService.js	Resend client, recipient builder (Solo/Doubles/Team), HTML template, async
server/services/pdfService.js	Server-side jsPDF A4 size pass generator with full team roster
server/config/env.js	RESEND_API_KEY, RESEND_FROM_EMAIL, APP_URL config properties
server/config/dbInit.js	email_status, email_error, email_sent_at columns auto-migration
server/controllers/registrationController.js	Email trigger after DB save + getRegistrationPassPDF controller
server/routes/publicRoutes.js	GET /api/public/registration-pass/:id route registered

11 Environment Variables — Render / Vercel par Set Karna Hai

&p Sirf yahi 3 email variables set karne hain — baaki sab already configured hai

Variable	Value
RESEND_API_KEY	re_g8Zm4LXm_CkPvb9nv67gdv7WW95a35SR5
RESEND_FROM_EMAIL	MPGI Sports <noreply@mpgisports.in>
APP_URL	https://mpgisports.in

Render par: Service ! Environment tab ! Add Environment Variable ! Save ! Auto-redeploy
Vercel par: Project ! Settings ! Environment Variables ! Add ! Save ! Redeploy

12 Domain & Infrastructure Setup Status

Service	Detail	Status
Resend Domain	mpgisports.in — SPF/DKIM verified	' Verified
ImprovMX Forwarding	noreply@ ! sports@mpgi.edu.in	' Active
MX Records	mx1 + mx2.improvmx.com set in Hostinger DNS	' Set
Git Branch	ankithome pushed to GitHub — main untouched	' Pushed
Gravatar	MPGI SPORTS APEX logo uploaded	' Done
Production Ready	All tests passed — ready to deploy	' 100 %

13 Live Test Results

Test	Result
Solo recipient extraction (1 email)	' PASS
Doubles deduplication (2 distinct emails)	' PASS
Team recipient extraction (4 valid emails, empty filtered)	' PASS
HTML template rendering (receipt ID, branding, pass link)	' PASS
Server-side PDF buffer (15,014 bytes, %PDF- signature)	' PASS
Resend live API dispatch (Message ID confirmed)	' PASS
PDF HTTP stream (Content-Type: application/pdf)	' PASS
Domain send from noreply@mpgisports.in	' PASS
14 game emails to ankit738885@gmail.com	' 14 / 14 DELIVERED
Content-Disposition: attachment (direct download)	' PASS

